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"Multimodal AI System for Ovarian Cancer Detection Using DNABERT and Swin Transformer"

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CERTIFICATE

This is to certify that the project work titled **"Multimodal AI System for Ovarian Cancer Detection Using DNABERT and Swin Transformer"** is carried out by **Nagasamhita - 23BTRCL059, Harshini- 23BTRCL215, Mounika - 23BTRCL216** bonafide student(s) of Bachelor of Technology at the School of Engineering & Technology, Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2023-2027**.

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Signature of Student(s)

ABSTRACT

The development of ovarian cancer leads to its most dangerous form because patients do not show symptoms until they reach advanced stages. Patients who receive treatment at earlier stages of their disease have better chances of survival but standard diagnostic methods which include histopathological examination and biomarker-based screening face challenges. The research demonstrates a multimodal deep learning framework which combines histopathological images with genomic data to enhance ovarian cancer detection accuracy. The system uses a Swin Transformer to extract image features at multiple levels while a BERT-based genomic encoder establishes gene expression relationships. The system uses an attention-based fusion mechanism to combine both modalities which improves its ability to classify data correctly.

The experimental results showed that the proposed system achieved better performance results than traditional CNN based systems because it achieved higher accuracy and better resilience. Multimodal transformer architectures prove effective because they enable systems to gather essential structural and molecular data which supports clinical decision-making processes.

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CHAPTER 1

Introduction

1.1 Introduction

Ovarian cancer continues to be one of the most critical challenges in women's healthcare due to its asymptomatic nature and late-stage detection. Unlike many other cancers, ovarian cancer often progresses silently, with symptoms appearing only in advanced stages. As a result, a significant proportion of cases are diagnosed at a point where treatment options become limited and less effective. Clinical studies indicate that early-stage detection can dramatically improve survival rates, whereas late-stage diagnosis is associated with significantly higher mortality. This highlights the urgent need for reliable, accurate, and early diagnostic systems.

The complexity of ovarian cancer further complicates the diagnostic process. The disease exhibits high heterogeneity in both morphological and genetic characteristics, meaning that tumor appearance and behavior can vary significantly across patients. This variability makes it difficult to establish consistent diagnostic criteria, even for experienced clinicians. Consequently, there is a growing demand for computational approaches that can assist in identifying subtle and complex patterns associated with the disease.

Traditional diagnostic methods rely on imaging techniques such as ultrasound, computed tomography (CT), and magnetic resonance imaging (MRI), along with histopathological examination of tissue samples. While histopathology remains the gold standard for diagnosis, it is inherently dependent on expert interpretation. This introduces challenges such as inter-observer variability, where different specialists may arrive at different conclusions when examining the same sample. Additionally, manual analysis is time-intensive and may not scale effectively in clinical settings with high patient volumes.

Biomarker-based detection methods, such as CA-125 testing, have also been used to support diagnosis. However, these approaches often lack sufficient sensitivity and specificity, particularly in early stages of the disease. False positives and false negatives remain a concern, limiting their reliability as standalone diagnostic tools. These limitations emphasize the need for more advanced, automated, and objective systems that can enhance diagnostic accuracy and consistency.

In recent years, artificial intelligence has emerged as a powerful tool in healthcare, particularly in medical image analysis and disease prediction. Machine learning techniques have been applied to clinical data to assist in risk assessment and classification tasks. However, early machine learning models relied heavily on manually engineered features and were limited in their ability to process complex and high-dimensional data such as images and genomic sequences.

The introduction of deep learning marked a significant advancement in this domain. Convolutional Neural Networks (CNNs) have been widely used for analyzing medical images and have

demonstrated strong performance in tumor detection and classification. These models are capable of automatically learning feature representations from raw image data, reducing the need for manual feature engineering. Despite their success, CNN-based models have inherent limitations. Their reliance on localized convolutional operations restricts their ability to capture long-range dependencies and global contextual information within images. This limitation is particularly relevant in histopathological analysis, where important diagnostic features may span across multiple regions.

Transformer-based architectures represent a significant shift in deep learning methodologies. Originally developed for natural language processing, transformers utilize self-attention mechanisms to model relationships between different parts of the input data. This allows them to capture both local and global dependencies more effectively than traditional convolutional approaches. Vision Transformers (ViT) have demonstrated the potential of this approach in image classification tasks; however, their high computational complexity limits their applicability in high-resolution medical imaging.

To address these challenges, the Swin Transformer architecture was introduced. It employs a hierarchical structure and a shifted window-based self-attention mechanism, enabling efficient computation while maintaining the ability to model global context. This makes it particularly suitable for medical image analysis, where both fine-grained details and large-scale structural patterns are important for accurate diagnosis.

While imaging provides valuable structural information, cancer is fundamentally a genetic disease driven by molecular-level changes. Genomic data plays a crucial role in understanding disease progression, identifying risk factors, and enabling personalized treatment strategies. Advances in sequencing technologies have made it possible to analyze DNA sequences at scale, but extracting meaningful insights from such data requires sophisticated computational models.

Transformer-based models such as DNABERT have demonstrated strong capabilities in genomic sequence analysis. By treating DNA sequences as a language and applying k-mer tokenization, these models can capture contextual relationships between nucleotide patterns. This enables the extraction of biologically relevant features that can support disease prediction and classification.

Despite advancements in both imaging and genomic analysis, most existing diagnostic systems rely on a single modality. Imaging-based models focus on structural features, while genomic models capture molecular information. However, relying on a single data source may lead to incomplete understanding of the disease. Ovarian cancer diagnosis requires integration of both morphological and molecular information to achieve higher accuracy and reliability.

Multimodal learning provides a solution to this limitation by combining multiple data sources into a unified framework. By integrating imaging and genomic data, multimodal systems can leverage complementary information, leading to improved diagnostic performance. This approach enables the model to capture a more comprehensive representation of disease characteristics.

1.2 Motivation

The motivation of this work is to develop a robust and scalable diagnostic framework that integrates advanced transformer-based architectures for both imaging and genomic data. The proposed system utilizes the Swin Transformer model for extracting hierarchical features from histopathological images and incorporates a genomic feature extraction pipeline inspired by DNABERT. A multimodal fusion strategy is then employed to combine these features into a unified representation for classification.

1.3 Aim and Objectives of the work

The primary contributions of this study include the development of a transformer-based image classification model, the implementation of a genomic feature extraction pipeline, and the design of a multimodal fusion architecture that integrates both modalities. The system is evaluated using standard performance metrics to assess its effectiveness in distinguishing between healthy and diseased samples.

The remainder of this report is organized as follows. The next section presents a detailed review of existing literature related to ovarian cancer detection using artificial intelligence. This is followed by a comprehensive description of the proposed methodology and model architecture. The training strategy and optimization techniques are then discussed, followed by the presentation of experimental results and performance evaluation. Finally, the report concludes with a discussion of findings and directions for future research.

CHAPTER 2

Literature Survey

2.1 Introduction

Ovarian cancer remains one of the leading causes of cancer-related mortality among women worldwide due to its asymptomatic nature and late-stage diagnosis. Early detection significantly improves survival rates; however, conventional diagnostic methods often fail to identify the disease at an early stage. In recent years, Artificial Intelligence (AI) has emerged as a powerful tool in medical diagnosis, particularly in cancer detection. This section presents a detailed review of existing research in ovarian cancer detection using machine learning, deep learning, transformer-based models, genomic analysis, and multimodal approaches.

2.2 Machine Learning-Based Approaches

Early studies in ovarian cancer detection relied on classical machine learning algorithms trained on structured clinical data.

Akazawa and Hashimoto (2020) applied machine learning techniques to clinical datasets for ovarian cancer diagnosis and demonstrated that models such as XGBoost and Random Forest achieved accuracy levels around 80%. Their study highlighted that machine learning could improve diagnostic consistency compared to manual assessment.

Similarly, Sadeghi et al. (2024) conducted a comprehensive review of deep learning and machine learning approaches in ovarian cancer diagnosis. Their findings indicated that traditional machine learning models perform well when trained on structured datasets but struggle with high-dimensional and unstructured data such as images and genomic sequences.

Other researchers have used Support Vector Machines (SVM) and Logistic Regression models for classifying benign and malignant tumors using clinical and biomarker data such as CA-125. While these approaches showed moderate success, their reliance on handcrafted features limits their scalability and generalization.

Overall, machine learning models provide a strong baseline but are insufficient for capturing complex patterns in medical imaging and genomic data.

2.3 Deep Learning in Medical Imaging

The introduction of deep learning significantly improved the performance of automated diagnostic systems. Convolutional Neural Networks (CNNs) have been widely used for analyzing medical images, including histopathological slides, CT scans, and ultrasound images.

Esteva et al. (2017) demonstrated that deep CNNs can achieve dermatologist-level accuracy in skin cancer classification, establishing the potential of deep learning in medical imaging. Inspired by such success, researchers applied CNN architectures such as ResNet, VGG, and EfficientNet to ovarian cancer detection.

Studies have reported that CNN-based models can achieve accuracy exceeding 90% in tumor classification tasks. These models automatically learn hierarchical feature representations, enabling the detection of subtle patterns such as irregular tissue morphology and tumor boundaries.

However, CNNs have inherent limitations. Their local receptive fields restrict their ability to capture longrange dependencies and global context. In histopathological images, important diagnostic features may be distributed across large regions, which CNNs may fail to fully capture.

Additionally, deeper CNN architectures increase computational complexity and may lead to overfitting, especially when training data is limited.

2.4 Transformer-Based Models in Medical Imaging

Transformer architectures have recently gained attention in computer vision tasks due to their ability to model global dependencies using self-attention mechanisms.

Dosovitskiy et al. (2020) introduced the Vision Transformer (ViT), which applies transformer architecture directly to image patches. ViT demonstrated superior performance compared to CNNs in large-scale image classification tasks.

In the medical domain, transformer-based models have shown promising results. Chen et al. (2021) applied transformer architectures for medical image segmentation and classification, achieving improved performance over traditional CNN-based approaches.

Transformers enable the model to capture relationships between distant regions of an image, which is particularly important in histopathological analysis. However, standard transformer models suffer from quadratic computational complexity with respect to image size, making them less efficient for highresolution medical images.

2.5 Swin Transformer and Its Advantages

To address the limitations of traditional transformers, Liu et al. (2021) introduced the Swin Transformer. This architecture uses a hierarchical structure and shifted window-based self-attention to reduce computational complexity while maintaining strong performance.

The Swin Transformer divides images into non-overlapping windows and computes self-attention within each window. By shifting the windows between layers, the model enables cross-window interaction, effectively capturing global context.

Recent studies have applied Swin Transformer to medical imaging tasks, including cancer detection. These models have demonstrated improved accuracy, precision, and recall compared to CNN-based and ViT-based models.

The hierarchical design of Swin Transformer makes it particularly suitable for histopathological image analysis, where both fine-grained textures and global structures are important.

2.6 Hybrid CNN–Transformer Models

To leverage the strengths of both CNNs and transformers, researchers have proposed hybrid architectures that combine local feature extraction with global attention mechanisms.

For example, hybrid models integrating EfficientNet with transformer modules have shown improved performance in medical image classification tasks. These models use convolutional layers to capture local features and transformer layers to model global dependencies.

Studies report that hybrid architectures achieve better generalization and higher accuracy compared to standalone CNN or transformer models. This demonstrates the importance of combining complementary feature extraction mechanisms.

2.7 Genomic Analysis and DNABERT

Cancer is fundamentally a genetic disease, and genomic analysis plays a crucial role in understanding disease progression. Advances in sequencing technologies have enabled large-scale analysis of DNA sequences, leading to the identification of mutations associated with ovarian cancer.

Ji et al. (2021) introduced DNABERT, a transformer-based model for genomic sequence analysis. DNABERT treats DNA sequences as a language and uses k-mer tokenization to learn contextual relationships between nucleotide patterns.

This approach enables the model to capture long-range dependencies in genomic data, which are essential for identifying biologically relevant patterns. DNABERT has been successfully applied in various genomic tasks, including gene prediction and disease classification.

Despite its effectiveness, genomic analysis is often performed independently of imaging-based systems, limiting its diagnostic potential.

2.8 Multimodal Learning in Medical AI

Multimodal learning integrates multiple data sources, such as imaging, genomic, and clinical data, to improve diagnostic performance.

Huang et al. (2020) demonstrated that combining imaging and clinical data improves cancer prediction accuracy compared to single-modality models. Similarly, studies in breast and lung cancer detection have shown that multimodal systems outperform traditional approaches.

Multimodal models capture complementary information, enabling a more comprehensive understanding of disease characteristics. However, integrating different data modalities presents challenges, including data alignment, feature fusion, and computational complexity.

Recent research has explored transformer-based multimodal architectures, but their application in ovarian cancer detection remains limited.

2.9 Research Gap

Based on the reviewed literature, several limitations can be identified:

- Most existing models rely on a single modality (either imaging or genomic data)
- CNN-based models fail to capture long-range dependencies
- Standard transformers are computationally expensive for medical images
- Limited research exists on transformer-based multimodal systems for ovarian cancer

These gaps highlight the need for a unified framework that integrates both structural and molecular information using efficient transformer-based architectures.

2.10 Summary

The literature indicates that while significant progress has been made in applying AI to ovarian cancer detection, existing approaches still face limitations in terms of feature representation and data integration. Multimodal transformer-based systems offer a promising solution by combining the strengths of imaging and genomic analysis. The proposed work addresses these challenges by integrating Swin Transformer for image analysis and DNABERT-based genomic feature extraction within a unified multimodal framework.

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

3.1 Introduction

The Software Requirement Specification (SRS) document provides a detailed description of the requirements for the proposed Multimodal AI System for Ovarian Cancer Detection. It outlines the functional and non-functional requirements, system constraints, and overall architecture of the system.

This system integrates CT imaging data and genomic expression data to perform accurate classification of ovarian cancer using advanced deep learning models such as Swin Transformer and DNABERT.

The SRS serves as a reference for developers, stakeholders, and researchers to understand system functionality and design.

3.1.1 Project Purpose

The primary purpose of this project is to develop an intelligent diagnostic system that:

- Detects ovarian cancer at an early stage
- Improves diagnostic accuracy by combining imaging + genomic data
- Reduces dependency on manual clinical interpretation
- Assists clinicians in decision-making with AI-based predictions

The system leverages multimodal learning to overcome limitations of single-modality approaches, as discussed in your report (integration of CT and genomic features).

3.2 Functional Requirements

The system must perform the following functions:

Input Functions

- Upload CT scan data in DICOM format
- Upload genomic expression data (TSV files)

Processing Functions

- Preprocess CT images (HU conversion, slice selection, normalization)
- Preprocess genomic data (gene selection, normalization)
- Extract features using:
    - Swin Transformer (image encoder)
    - DNABERT-inspired model (genomic encoder)

Core System Functions

- Perform multimodal feature fusion
- Run classification (Tumor / Healthy)

- Generate prediction with confidence score

Output Functions

- Display results to the user
- Provide probability scores
- Generate clinical report (PDF export)

These functionalities align with the workflow described in your system diagrams (Use Case & Sequence diagrams in Chapter 4).

3.3 Non-Functional Requirements

Performance Requirements

- System should provide predictions within a reasonable time (< few seconds per sample)
- Must support large datasets (medical images + genomic data)

Reliability

- High accuracy and consistency in predictions
- System should handle incomplete or noisy data gracefully

Scalability

- Capable of handling increasing dataset sizes
- Extendable to other cancer types or modalities

3.3.1 Safety Requirements

- The system must not replace clinical diagnosis but act as a decision-support tool
- Incorrect predictions (especially false negatives) must be minimized
- Clear disclaimers should be included in reports

As highlighted in your evaluation section, false negatives are critical in medical systems.

3.3.2 Security Requirements

- Secure handling of patient data
- Data privacy must be maintained (no unauthorized access)
- Encryption of sensitive medical data
- Role-based access control for users

3.3.3 Software Quality Attributes

- Accuracy: High classification performance (AUC, F1-score)
- Usability: Easy-to-use interface (e.g., Streamlit UI mentioned in your workflow)
- Maintainability: Modular architecture (image encoder, genomic encoder, fusion model)
- Robustness: Ability to handle missing or inconsistent data
- Efficiency: Optimized deep learning inference

3.4 System Requirements

3.4.1 Software Requirements

- Programming Language: Python
- Frameworks/Libraries:
    - PyTorch / TensorFlow
    - HuggingFace Transformers
    - OpenCV / PIL
    - NumPy, Pandas
    - Scikit-learn
- Development Environment:

  - Google Colab / Jupyter Notebook
- UI Framework:
  - Streamlit

These align with your implementation section using Colab and deep learning frameworks.

3.4.2 Hardware Requirements

- Processor: Intel i5 or higher
- RAM: Minimum 8GB (16GB recommended)
- GPU: NVIDIA GPU (e.g., T4 for training, as used in your project)
- Storage: Minimum 50GB for datasets

Your report mentions training on **Google Colab with NVIDIA T4 GPU**.

3.5 Analysis Models

The system design is represented using UML diagrams to define structure and behavior:

- **Use Case Diagram** – Shows interaction between users (Clinician, Radiologist, Analyst) and system
- **Class Diagram** – Represents system components like:
  - DICOMDataset
  - GenomicDataset
  - SwinImageEncoder
  - GenomicBERTEncoder
  - Fusion Classifier
- **Sequence Diagram** – Describes workflow from input → processing → prediction
- **Activity Diagram** – Shows parallel processing of CT and genomic pipelines

- **State Diagram** – Represents system states (Idle → Upload → Processing → Prediction → Report)

CHAPTER 4

DESIGN & MODELLING OF SYSTEM

Unified Modelling Language (UML) is analogous to the blueprints used in other fields and consists of different types of diagrams. In the aggregate, UML diagrams describe the boundary, structure, and behaviour of the system and the objects within it. For this multi-modal cancer detection system — integrating Swin Transformer-based CT imaging analysis with BERT-based genomic expression modelling — the following UML diagrams have been designed:

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram
- State Diagram

4.1 Use Case Diagram

A use case diagram describes a set of actions that the system performs in collaboration with one or more external users (actors). Each use case provides an observable and valuable result to the actors or other stakeholders of the system. The actors identified in this multi-modal cancer detection system are: Clinician, Radiologist, and Genomics Analyst.

The primary use cases are: Upload CT DICOM Scan, Upload Genomic Expression File, Preprocess CT Slices (HU windowing, pelvic slice filtering, axial series selection), Preprocess Genomic Data (TPM normalization, top-N gene selection), Run Multi-Modal Fusion Inference (Swin + BERT), View Prediction with Confidence Score, and Export Clinical Report. The system orchestrates all internal preprocessing and inference steps automatically upon input submission.

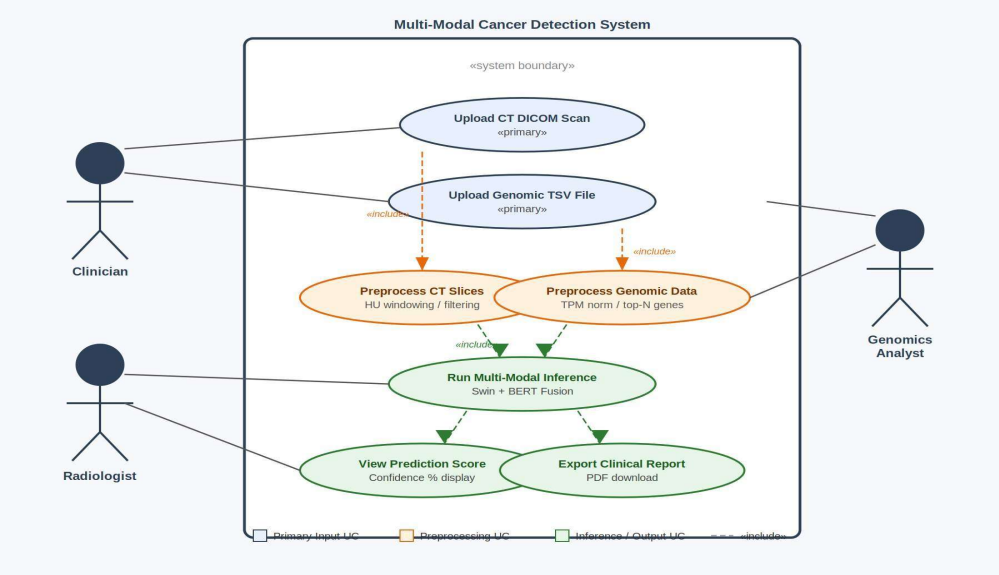


Fig 4.1: Use Case Diagram of the Multi-Modal Cancer Detection System

4.2 Class Diagram

A class diagram illustrates the relationships and source code dependencies among classes in UML. It defines the methods and variables in each object that is a specific entity in the program. The multi-modal system is organized around four major class groups: data layer, encoder modules, fusion classifier, and training engine.

The core classes include: DICOMDataset (handles DICOM loading, HU conversion, pelvic filtering, and slice sampling), GenomicDataset (reads TSV expression files, filters top-N variable genes, applies StandardScaler normalization), SwinImageEncoder (wraps the Swin Transformer backbone with a learned projection head), GenomicBERTEncoder (implements a lightweight BERT architecture for gene-expression token sequences), MultiModalFusionClassifier (concatenates both encoder embeddings and applies MLP fusion layers to produce the final binary prediction), and TrainingEngine (manages the optimization loop, early stopping, and checkpoint saving).

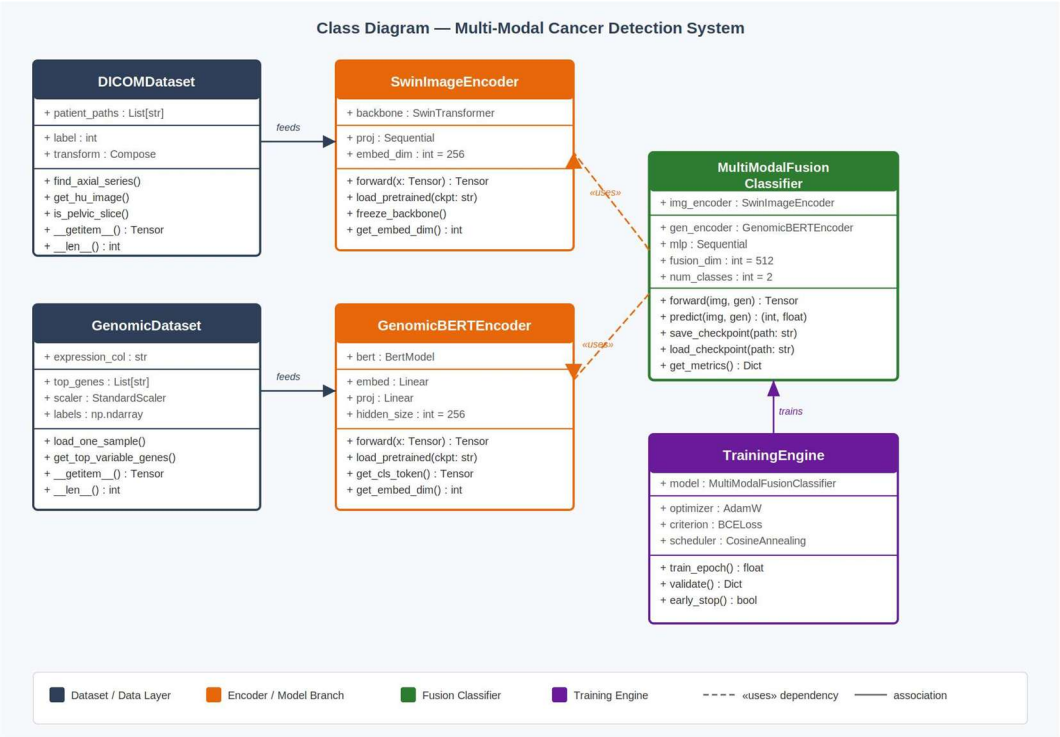


Fig 4.2: Class Diagram showing module dependencies and data flow

4.3 Sequence Diagram

A sequence diagram shows, as parallel vertical lines (lifelines), different processes or objects that live simultaneously, and as horizontal arrows the messages exchanged between them in the order they occur. The diagram below captures the temporal flow of interactions during a complete inference session.

The flow proceeds as follows:

(1) The Clinician uploads a DICOM folder and a genomic TSV file via the Streamlit UI.

(2) The Preprocessor extracts axial slices, filters pelvic regions by Hounsfield Unit thresholds, and applies windowing.

(3) Simultaneously, the Genomic branch selects the top-N most variable genes and standardizes expression values.

(4) Both tensors are dispatched in parallel to SwinEncoder and BERTEncoder.

(5) The FusionClassifier receives the concatenated embeddings and returns softmax probabilities.

(6) The UI renders the predicted class and confidence percentage.

(7) On request, the system generates and returns a PDF clinical report.

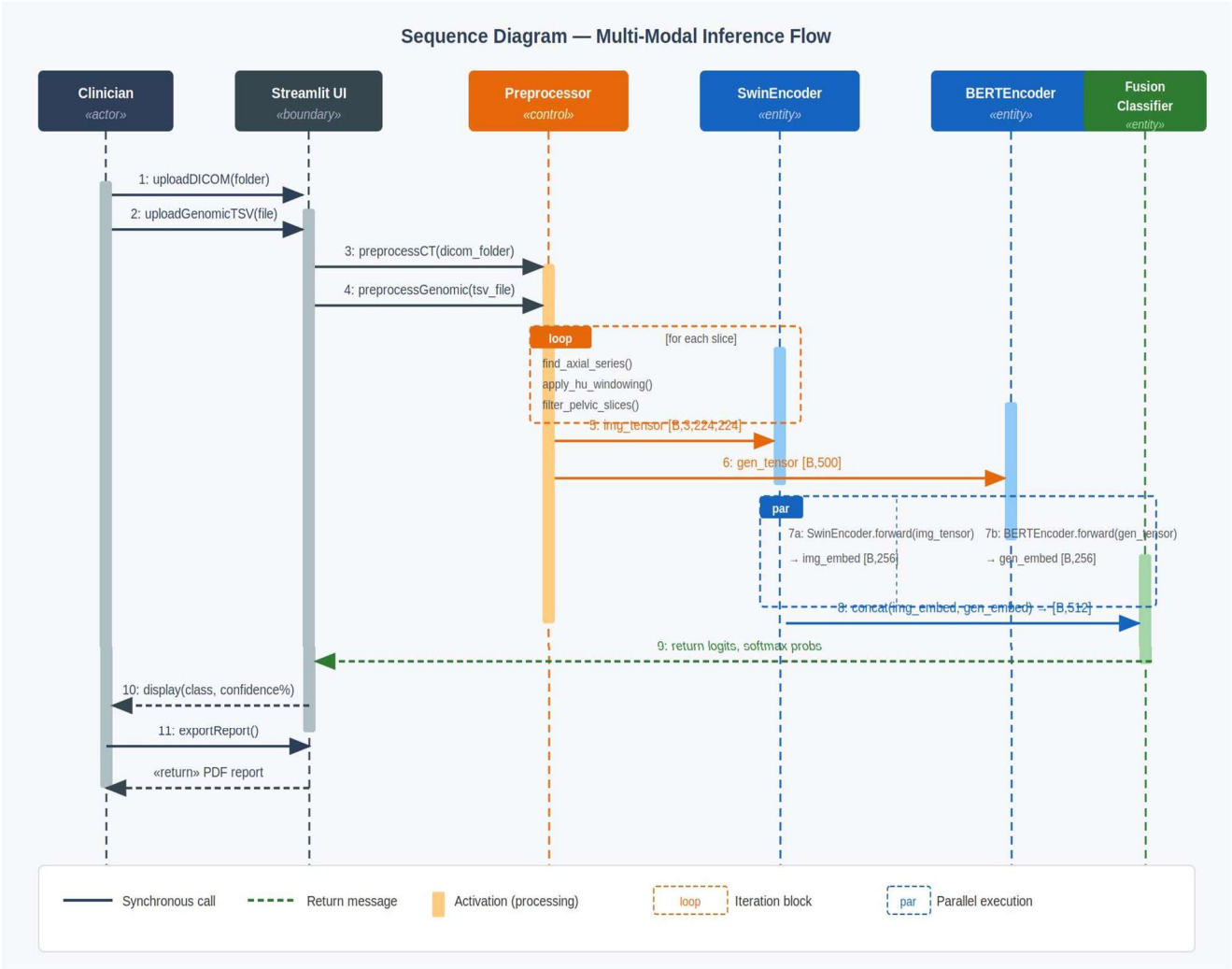


Fig 4.3: Sequence Diagram illustrating inference-time message flow

4.4 Activity Diagram

An activity diagram is a flowchart that represents the flow from one activity to another. It describes the dynamic aspects of the system and is used here to model the complete dual-modality processing workflow — from raw data ingestion through to the final prediction output.

The activity flow is organized across two parallel swim lanes — the CT Imaging Pipeline and the Genomic Expression Pipeline. In the CT lane, raw DICOM files are loaded, axial series are identified and sorted by slice count, Hounsfield Unit conversion is applied, pelvic slices are filtered by a tissue-to-bone ratio threshold, and the result is resampled to a fixed 64-slice volume before normalization and augmentation. In the genomic lane, TSV files are parsed, rows prefixed with 'N_' are discarded, TPM values are extracted, gene variance is computed, and the top-500 genes are retained, normalized, and class-balanced. Both swim lanes converge at a JOIN bar that feeds into the Fusion Classifier, which produces the final Tumor/Healthy prediction.

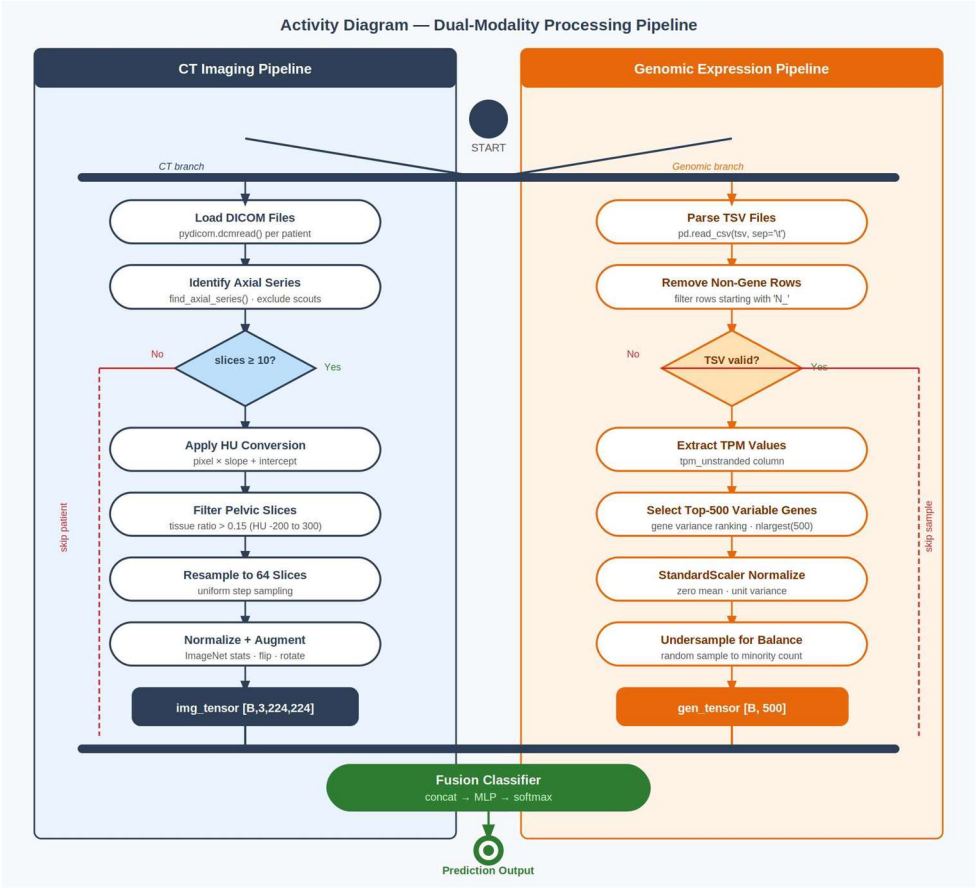


Fig 4.4: Activity Diagram of the dual-modality processing pipeline

4.5 State Diagram

A state diagram, sometimes known as a state machine diagram, is a type of behavioural diagram in UML that shows transitions between various objects. Here it models the discrete states of the application from the user's perspective throughout the session lifecycle.

The system transitions through the following states: Idle (awaiting user inputs), CT Upload In Progress (validating DICOM format), Genomic Upload In Progress (validating TSV schema), Preprocessing (both modalities active in parallel — HU windowing, gene normalization), Model Inference Running (Swin + BERT forward pass, MLP fusion), Prediction Ready (displaying result with confidence score and ROC overlay), Export Report (generating PDF), and Session Ended. Guard conditions protect each transition — for instance, the system moves from Preprocessing to Inference only when both branches complete without error. Any exception transitions the system to an Error State, from which the user can retry and reset. A self-loop on Prediction Ready allows the clinician to initiate a new upload without ending the session.

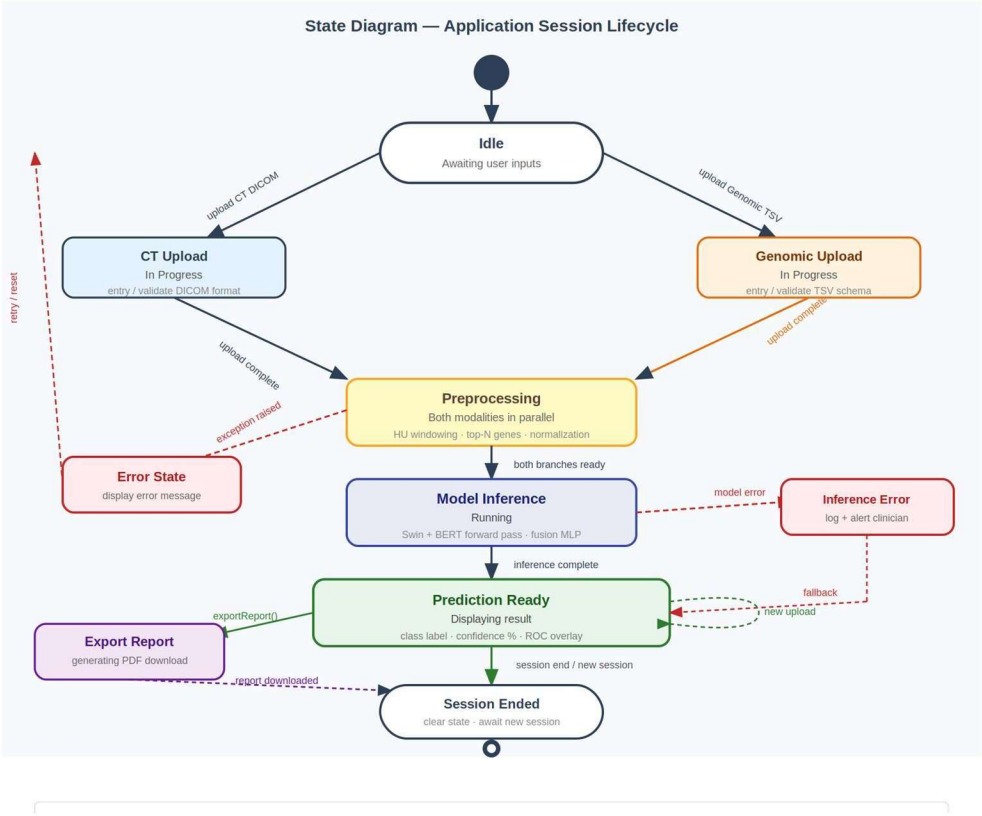


Fig 4.5: State Diagram of the application session lifecycle

CHAPTER 5

ALGORITHM AND METHODS

This chapter presents a comprehensive account of the algorithmic foundations underlying the proposed multi-modal cancer detection system. The system is designed to simultaneously analyze two distinct biological data streams — CT imaging data captured in DICOM format and RNA-seq gene expression profiles stored in TSV format — and fuse them through a deep learning architecture for binary classification of ovarian cancer (Tumor vs. Healthy). Each section below describes in detail the strategies employed for data acquisition, data preprocessing, and the design of the proposed deep neural network.

The primary motivation for combining imaging and genomic modalities is that each carries complementary diagnostic information. CT scans reveal macroscopic morphological features — tumor shape, density, location, and size — while genomic expression profiles expose the underlying molecular dysregulation that often precedes or accompanies visible structural changes. A system that leverages both is inherently more robust and clinically meaningful than any single-modality approach.

5.1 Data Acquisition

Data acquisition is the foundational step of any machine learning pipeline. The quality, diversity, and representativeness of the acquired data directly determine the upper bound of any model's performance. For this system, two independent datasets were acquired and curated — one for each modality.

5.1.1 CT Imaging Dataset

CT scans were stored in DICOM (Digital Imaging and Communications in Medicine) format, the globally accepted standard for medical imaging. Each patient's data was organized as a folder containing one or more series of axial CT slices, each encoded as an individual .dcm file. The imaging dataset comprised both healthy control patients and cancer-positive (diseased) cases, reflecting a real-world clinical distribution.

A fundamental challenge with raw DICOM archives is their heterogeneity. A single patient folder may contain multiple series — axial, coronal, sagittal, scout/localizer images, and reformatted MPR sequences. Including non-axial or scout images would corrupt the spatial coherence of the dataset and introduce misleading visual patterns. To address this, a robust automated axial series selection procedure was implemented:

```python
def find_axial_series(patient_dir):
    all_dcm = glob.glob(
        os.path.join(patient_dir, '**', '*.dcm'),
        recursive=True
    )

    series_dict = {}

    for dcm_path in all_dcm:
        try:
            ds = pydicom.dcmread(dcm_path, stop_before_pixels=True)

            series_uid = str(getattr(ds, 'SeriesInstanceUID', 'unknown'))

            image_type = list(getattr(ds, 'ImageType', []))
            if 'LOCALIZER' in image_type or 'SCOUT' in image_type:
                continue

            desc = str(getattr(ds, 'SeriesDescription', '')).upper()

            # Skip non-axial
            if any(x in desc for x in ['CORONAL', 'SAGITTAL', 'MPR', 'REFORMAT']):
                continue

            if series_uid not in series_dict:
                series_dict[series_uid] = []

            series_dict[series_uid].append(dcm_path)

        except:
            continue

    if not series_dict:
        return all_dcm

    best_series = max(series_dict.values(), key=len)
    return sorted(best_series)
```

Fig 5.1.1: find_axial_series()

This function reads only the DICOM header (stop_before_pixels=True), which avoids loading large pixel arrays during the selection pass. It then filters out scout and localizer images using the ImageType tag, excludes coronal, sagittal, MPR, and reformatted series using the SeriesDescription tag, groups remaining files by SeriesInstanceUID, and selects the series with the highest slice count as the best axial candidate.

Following series selection, a minimum quality gate was enforced: only patients with at least 10 valid axial slices and at least 5 anatomically relevant pelvic slices were retained. Patients below these thresholds were silently skipped. This gate eliminates incomplete scans, corrupted archives, and edge-case acquisitions that would otherwise act as noise in the training distribution.

For the diseased cohort, all qualifying DICOM files within each patient folder were included directly, as the disease dataset had been previously curated to contain predominantly pelvic CT

series. For the healthy cohort, the additional pelvic anatomical filtering step (described in Section 5.2) was applied to ensure comparability between the two class distributions.

5.1.2 Genomic Expression Dataset

RNA-seq gene expression data were structured following the TCGA (The Cancer Genome Atlas) STAR pipeline format. Each patient sample was stored as a separate subdirectory containing a single TSV file. The TSV file contained columns for gene_id, unstranded read counts, FPKM (Fragments Per Kilobase Million), FPKM-UQ, and TPM (Transcripts Per Million) values.

TPM was selected as the primary expression metric for the following reasons. Unlike raw read counts, TPM normalizes both for gene length (dividing by gene length in kilobases) and for total sequencing depth (dividing by the sum of all length-normalized values and scaling to one million). This double normalization ensures that TPM values are directly comparable across samples regardless of sequencing depth or transcript length distribution — a critical property when training a machine learning model on a multi-patient cohort.

Dataset discovery was performed by scanning the root directories for tumor and healthy sample subdirectories:

```
DATASET_ROOT = "/content/genomic data set"
TUMOR_DIR    = os.path.join(DATASET_ROOT, "tumor")
HEALTHY_DIR  = os.path.join(DATASET_ROOT, "healthy")

# Sanity check
tumor_samples   = [d for d in glob.glob(os.path.join(TUMOR_DIR,   "*")) if os.path.isdir(d)]
healthy_samples = [d for d in glob.glob(os.path.join(HEALTHY_DIR, "*")) if os.path.isdir(d)]
print(f"Tumor  sample folders: {len(tumor_samples)}")
print(f"Healthy sample folders: {len(healthy_samples)}")
```

Fig 5.1.2: Dataset Root Path Setup

Each sample folder was loaded using a dedicated function that enforces strict schema validation — rows whose gene_id begins with "N_" (indicating non-gene alignment statistics such as N_unmapped and N_multimapping) were filtered out, as these represent alignment summary rows rather than actual gene expression measurements:

```
def load_one_sample(sample_dir, expression_col):
    tsv_files = glob.glob(os.path.join(sample_dir, "*.tsv"))
    if not tsv_files:
        raise FileNotFoundError(f"No TSV found in {sample_dir}")
    df = pd.read_csv(tsv_files[0], sep="\t", comment="#", low_memory=False)
    df = df[~df["gene_id"].str.startswith("N_", na=False)]
    df = df[df["gene_id"] != "gene_id"]
    df = df.set_index("gene_id")
    expr = pd.to_numeric(df[expression_col], errors="coerce").fillna(0.0)
    return expr
```

Fig 5.1.3: load_one_sample()

Because the full human genome contains over 60,000 annotated gene IDs, feeding the raw expression vector of this dimensionality into a model directly is computationally prohibitive and statistically problematic (due to the curse of dimensionality). A feature selection step was therefore applied: the top-500 most variably expressed genes were identified across a stratified random sample of 60 patients (30 tumor, 30 healthy) and retained as the fixed gene panel for all subsequent sample loading.

Variance-based gene selection is a standard dimensionality reduction technique in transcriptomics. Genes with near-zero variance across a patient cohort carry no discriminative signal for classification and only add noise. By retaining the 500 highest-variance genes, the representation is compact, memory-efficient, and biologically meaningful — high-variance genes are typically those actively regulated or dysregulated between biological conditions.

Parameter	**CT Imaging Dataset**	**Genomic Dataset**
Input Format	DICOM (.dcm)	TSV (TCGA STAR pipeline)
Expression / Intensity Metric	Hounsfield Units (HU)	TPM (Transcripts Per Million)
Feature Dimensionality	64 slices × 224 × 224 px	Top 500 variable genes
Normalization Strategy	HU windowing + ImageNet stats	StandardScaler (μ=0, σ=1)
Class Balancing	Random undersampling	Random undersampling
Train / Val / Test Split	80 / 10 / 10 (stratified)	80 / 10 / 10 (stratified)

Table 5.1: Data Acquisition Parameters Across Both Modalities

5.2 Data Pre-processing

Raw data from both modalities required extensive preprocessing before being fed into the neural network. This section describes the complete transformation pipeline for each branch.

5.2.1 CT Image Pre-processing

Hounsfield Unit Conversion. Raw DICOM pixel values are stored in scanner-specific digital units. To convert them to the universally standardized Hounsfield Unit (HU) scale, the RescaleSlope and RescaleIntercept metadata fields embedded in each DICOM file header are applied.

On the HU scale, air is −1000, water is 0, soft tissue ranges from −100 to +100, and dense cortical bone lies above +400. This physical grounding means HU values are directly comparable across different scanners and acquisition protocols.

```
import numpy as np

def get_hu_image(dcm):
    img = dcm.pixel_array.astype(np.float32)

    slope = getattr(dcm, "RescaleSlope", 1)
    intercept = getattr(dcm, "RescaleIntercept", 0)

    return img * slope + intercept


def is_pelvic_slice(dcm):
    hu = get_hu_image(dcm)

    # Tissue + bone region
    mask = (hu > -200) & (hu < 300)

    ratio = np.sum(mask) / mask.size

    return ratio > 0.15
```

Fig 5.2.1: Conversion and Pelvic Slice Detection

Uniform step sampling preserves the full anatomical extent of the pelvic region rather than truncating from either end, ensuring that apical, mid-pelvic, and inferior pelvic structures are all represented in the sampled volume.

Spatial Normalization and Augmentation Each retained slice is resized to 224 × 224 pixels using bilinear interpolation to match the input resolution expected by the Swin Transformer pre-trained on ImageNet. Since DICOM pixel arrays are single-channel grayscale, each slice is replicated across three channels to form a compatible 3-channel tensor. The pixel values are then normalized using ImageNet statistics:

- Mean: [0.485, 0.456, 0.406]
- Standard Deviation: [0.229, 0.224, 0.225]

During training, the following stochastic augmentations are applied to improve generalization:

Augmentation	Parameter	Rationale
Random Horizontal Flip	p = 0.5	Patients may be scanned in variable orientations
Random Rotation	± 15°	Accounts for patient positioning variation

Augmentation	Parameter	Rationale
Colour Jitter (brightness)	factor = 0.2	Simulates scanner acquisition variability
Resize	224 × 224 px	Required input size for Swin Transformer
Normalize	ImageNet mean / std	Aligns with pre-training distribution

Table 5.2: CT Image Augmentation Parameters

No augmentation is applied to the validation or test sets — only resizing and normalization — to ensure unbiased performance evaluation.

5.2.2 Genomic Data Pre-processing

Gene Expression Loading. After top-N gene selection, each sample's expression vector is loaded by calling load_one_sample() and reindexing to the fixed gene panel. Missing values (genes absent from a particular sample's TSV) are filled with zero, reflecting the assumption of no detected expression rather than discarding the sample entirely.

Standardization. The aligned expression matrix across all samples is standardized using sklearn's StandardScaler, which subtracts the column-wise mean and divides by the column-wise standard deviation. This produces a zero-mean, unit-variance representation for each gene dimension independently:

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train).astype(np.float32)
X_val   = scaler.transform(X_val).astype(np.float32)
X_test  = scaler.transform(X_test).astype(np.float32)

print(f"Train: {len(y_train)} | Val: {len(y_val)} | Test: {len(y_test)}")
```

Fig 5.2.2: StandardScaler Normalization

A critical implementation detail is that the scaler is fit exclusively on the training set and then applied (transform only) to the validation and test sets. Fitting on the full dataset would constitute data leakage, artificially inflating validation and test performance metrics.

Class Balancing. A systematic class imbalance is common in clinical datasets — the healthy cohort typically outnumbers cancer-positive cases. To prevent the model from learning a trivial majority-class bias, random undersampling is applied to bring the majority class count down to match the minority class.

Dataset Splitting. Both modalities share an identical patient-level stratified 80/10/10 train/validation/test split to ensure that no patient's data appears in more than one partition:

```
X = np.log1p(X)

X_train, X_temp, y_train, y_temp = train_test_split(
    X, y,
    test_size=(CONFIG["test_size"] + CONFIG["val_size"]),
    stratify=y, random_state=CONFIG["seed"]
)
val_frac = CONFIG["val_size"] / (CONFIG["test_size"] + CONFIG["val_size"])
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp,
    test_size=(1 - val_frac),
    stratify=y_temp, random_state=CONFIG["seed"]
)
```

Fig 5.2.3: log1p Transform

Step	CT Imaging	Genomic Expression
1. Load	pydicom.dcmread()	pd.read_csv(tsv, sep='\t')
2. Convert	HU = pixel × slope + intercept	Select TPM column, coerce to float32
3. Filter	Pelvic slice tissue ratio > 0.15	Remove 'N_' prefixed gene IDs
4. Resample	Uniform step to 64 slices	Reindex to top-500 gene panel
5. Normalize	ImageNet mean/std per channel	StandardScaler (μ=0, σ=1) on train
6. Augment	Flip, rotation, jitter (train only)	None (tabular data)
7. Output Shape	[B, 3, 224, 224]	[B, 500] float32

Table 5.3: Complete Pre-processing Pipeline for Both Modalities

5.3 Proposed Network

The proposed system is a multi-modal deep learning architecture that independently encodes CT imaging and genomic expression data through two specialized branches, then combines their representations via a learned fusion mechanism for final classification. Each component is described in detail below.

5.3.1 Swin Transformer Image Encoder

The imaging branch of the network employs a Swin Transformer Tiny (swin_tiny_patch4_window7_224), loaded via the timm library with ImageNet-21K pre-trained weights.

The input image is divided into non-overlapping patches of size 4 × 4 pixels. The patch embedding projects each patch into a C-dimensional token. Through four successive Swin stages, the spatial resolution is progressively downsampled while the channel depth increases: Stage 1 (56×56, 96ch) → Stage 2 (28×28, 192ch) → Stage 3 (14×14, 384ch) → Stage 4 (7×7, 768ch). The final 768-dimensional feature vector is extracted from the global average pool of Stage 4's output.

The original classification head is removed and replaced with a custom projection head that reduces the 768-dimensional Swin feature vector to a 256-dimensional embedding:

```python
class SwinCTClassifier(nn.Module):

    def __init__(self, num_classes=2, dropout=0.3):
        super().__init__()
        # Load pretrained Swin-Tiny (same as your swin_model.ipynb)
        self.backbone = timm.create_model(
            'swin_tiny_patch4_window7_224',
            pretrained=True,
            num_classes=0   # remove head → returns 768-d features
        )
        feat_dim = self.backbone.num_features  # 768 for Swin-Tiny

        # Custom classification head
        self.head = nn.Sequential(
            nn.LayerNorm(feat_dim),
            nn.Dropout(dropout),
            nn.Linear(feat_dim, 256),
            nn.GELU(),
            nn.Dropout(dropout),
            nn.Linear(256, num_classes)
        )

    def forward(self, x):
        features = self.backbone(x)   # (B, 768)
        return self.head(features)    # (B, 2)


model = SwinCTClassifier(num_classes=2, dropout=0.3).to(device)

n_params = sum(p.numel() for p in model.parameters() if p.requires_grad)
print(f'Model ready — Trainable params: {n_params:,}')
```

Fig 5.3.1: SwinCTClassifier — Swin-Tiny Pretrained Backbone

LayerNorm is applied after the linear projection for training stability, and GELU activation introduces a smooth non-linearity prior to the fusion stage. During the early epochs of training, the

Swin backbone weights are optionally frozen to prevent catastrophic forgetting while the projection head warms up. The backbone is unfrozen in later epochs for joint fine-tuning.

5.3.2 Genomic BERT Encoder

The genomic branch employs a lightweight BERT (Bidirectional Encoder Representations from Transformers) model, configured from scratch using HuggingFace's BertConfig — not pre-trained on natural language, but initialized fresh and trained end-to-end on gene expression sequences. The rationale for applying a BERT architecture to gene expression data is as follows.

A gene expression profile of 500 values can be conceptualized as a sequence of 500 tokens, where each token represents one gene and its value encodes the expression level.

The embedded gene sequence is passed into the BERT encoder stack, which is configured with the following hyperparameters derived from ablation experiments:

Hyperparameter	Value	Justification
Hidden Size	256	Matches image encoder output dimension
Number of Attention Heads	8	32-dimensional attention per head
Number of Hidden Layers	4	Sufficient depth for 500-token sequences
Intermediate (FFN) Size	512	2× hidden size (standard BERT ratio)
Dropout Rate	0.35	Higher than standard BERT (0.1) to combat overfitting on small datasets
Maximum Sequence Length	500	Equal to top_n_genes

Table 5.4: Genomic BERT Encoder Hyperparameter Configuration

The [CLS] token at position 0 of the last hidden state is used as the aggregate representation of the entire gene expression sequence, following standard BERT classification practice. A linear projection maps the [CLS] embedding to the same 256-dimensional space as the imaging branch:

```python
class GenomicBERT(nn.Module):

    def __init__(self, n_genes: int, config: dict):
        super().__init__()
        H = config["hidden_size"]

        # Project each gene's scalar value → hidden_size vector
        self.gene_proj = nn.Linear(1, H)

        # Learnable positional embedding (one per gene position + CLS)
        self.pos_emb = nn.Embedding(n_genes + 1, H)

        self.norm = nn.LayerNorm(H)
        self.drop = nn.Dropout(config["dropout"])

        # BERT encoder
        bert_cfg = BertConfig(
            hidden_size=H,
            num_attention_heads=config["num_attention_heads"],
            num_hidden_layers=config["num_hidden_layers"],
            intermediate_size=config["intermediate_size"],
            max_position_embeddings=n_genes + 2,
            hidden_dropout_prob=config["dropout"],
            attention_probs_dropout_prob=config["dropout"],
            vocab_size=1,   # not used — we supply embeddings directly
        )
        self.encoder = BertModel(bert_cfg, add_pooling_layer=False)

        # Learnable CLS token (shared across all samples)
        self.cls_token = nn.Parameter(torch.zeros(1, 1, H))

        # Classification head
        self.head = nn.Sequential(
            nn.Linear(H, 128),
            nn.ReLU(),
            nn.Dropout(config["dropout"]),
            nn.Linear(128, 2)
        )
```

Fig 5.3.2: GenomicBERT — Full Architecture

5.3.3 Multi-Modal Fusion Classifier

The fusion classifier receives two 256-dimensional embeddings — one from the Swin encoder and one from the Genomic BERT encoder — and combines them into a joint 512-dimensional representation via concatenation. Concatenation is preferred over additive fusion or attention-weighted fusion because it preserves the full information content of both branches without imposing a prior assumption about their relative contribution. The model is free to learn the optimal weighting implicitly through the subsequent MLP layers.

The fusion MLP consists of two fully connected layers with Batch Normalization, GELU activation, and Dropout between them:

```python
class GenomicBERT(nn.Module):

    def __init__(self, n_genes: int, config: dict):
        super().__init__()
        H = config["genomic_hidden_size"]

        self.gene_proj = nn.Linear(1, H)
        self.pos_emb   = nn.Embedding(n_genes + 1, H)
        self.norm      = nn.LayerNorm(H)
        self.drop      = nn.Dropout(config["genomic_dropout"])

        bert_cfg = BertConfig(
            hidden_size              = H,
            num_attention_heads      = config["genomic_attn_heads"],
            num_hidden_layers        = config["genomic_layers"],
            intermediate_size        = config["genomic_intermediate"],
            max_position_embeddings  = n_genes + 2,
            hidden_dropout_prob      = config["genomic_dropout"],
            attention_probs_dropout_prob = config["genomic_dropout"],
            vocab_size               = 1,
        )
        self.encoder   = BertModel(bert_cfg, add_pooling_layer=False)
        self.cls_token = nn.Parameter(torch.zeros(1, 1, H))

        # standalone head (used when training genomic branch alone)
        self.head = nn.Sequential(
            nn.Linear(H, 128), nn.ReLU(),
            nn.Dropout(config["genomic_dropout"]),
            nn.Linear(128, config["num_classes"])
        )
```

```python
    def forward(self, x):
        seq  = self._embed(x)
        mask = torch.ones(seq.shape[0], seq.shape[1], device=x.device)
        out  = self.encoder(inputs_embeds=seq, attention_mask=mask)
        return self.head(out.last_hidden_state[:, 0, :])

    def get_embedding(self, x):
        """Returns CLS embedding (B, hidden_size) for fusion."""
        seq  = self._embed(x)
        mask = torch.ones(seq.shape[0], seq.shape[1], device=x.device)
        out  = self.encoder(inputs_embeds=seq, attention_mask=mask)
        return out.last_hidden_state[:, 0, :]          # (B, H)
```

Fig 5.3.3: GenomicBERT — BERT Encoder Architecture for Gene Expression Sequences

Fig 5.3.4: GenomicBERT forward() and get_embedding() — CLS Token Extraction for Fusion

Batch Normalization before activation accelerates convergence and reduces sensitivity to the learning rate. Dropout at p=0.3 provides regularization against overfitting, which is a significant risk given the relatively small size of curated medical datasets.

5.3.4 Loss Function and Optimization

The network is trained end-to-end using the following optimization setup:

Loss Function. Cross-Entropy Loss (nn.CrossEntropyLoss) is used as the primary objective. For a 2-class problem this is equivalent to Binary Cross-Entropy applied to the logit outputs. The loss function is defined as:

$$L = -[y \cdot \log(p) + (1-y) \cdot \log(1-p)]$$

where $y \in \{0, 1\}$ is the true class label and p is the predicted probability for the positive (Tumor) class. Cross-entropy penalizes confident wrong predictions far more heavily than uncertain ones, which is appropriate for a clinical screening context.

Early Stopping. Training is monitored on the validation set loss with a patience of 15 epochs. If validation loss does not improve for 15 consecutive epochs, training terminates and the best-performing checkpoint (lowest validation loss) is restored. This prevents both overfitting and unnecessary computation.

```
if vl_loss < best_val_loss:
    best_val_loss = vl_loss
    patience_count = 0
    torch.save({
        "model_state": fusion_model.state_dict(),
        "optimizer":   optimizer.state_dict(),
        "config":      CONFIG,
        "gene_ids":    gene_ids,
        "n_genes":     n_genes,
        "scaler":      scaler,
        "epoch":       epoch,
    }, CONFIG["FUSION_CKPT"])
    print(f"  ✓ Best model saved (val_loss={vl_loss:.4f})")
else:
    patience_count += 1
    if patience_count >= CONFIG["patience"]:
        print(f"\nEarly stopping at epoch {epoch} (no improvement for {CONFIG['patience']} epochs).")
        break
```

Fig 5.3.5: Best Model Checkpoint Saving and Early Stopping Logic — Fusion Training Loop

CHAPTER-6

IMPLEMENTATION AND TESTING

This chapter presents the step-by-step implementation of the multi-modal pipeline, covering environment setup, DICOM data acquisition from Google Drive, pelvic slice extraction, slice visualization, and data augmentation strategies. All experiments were conducted on Google Colab using an NVIDIA T4 GPU.

6.1 Data Acquisition

The raw CT datasets were stored as ZIP archives on Google Drive. After mounting the Drive, the healthy and diseased archives were extracted to local Colab paths:

```
import zipfile
import os

healthy_zip = "/content/drive/MyDrive/data/health.zip"
disease_zip = "/content/drive/MyDrive/data/disease.zip"

healthy_extract = "/content/Mydrive/healthy"
disease_extract = "/content/Mydrive/diseased"

# Extract healthy
if not os.path.exists(healthy_extract):
    with zipfile.ZipFile(healthy_zip, 'r') as zip_ref:
        zip_ref.extractall(healthy_extract)

# Extract disease
if not os.path.exists(disease_extract):
    with zipfile.ZipFile(disease_zip, 'r') as zip_ref:
        zip_ref.extractall(disease_extract)
```

Fig 6.1.1: Mounting Google Drive and Extracting CT Dataset

For each patient folder in the healthy cohort, the find_axial_series() function (Section 5.1) was called to retrieve the best axial series. Slices were then evaluated for pelvic anatomical content using is_pelvic_slice(). Patients with fewer than 20 qualifying pelvic slices were excluded. Retained slices were copied to the output directory:

The diseased dataset was processed using the same DICOM copy procedure. After processing, the total DICOM slice counts were verified:

```
def count_dicoms(path):
    total = 0
    for root, _, files in os.walk(path):
        total += len([f for f in files if f.endswith(".dcm")])
    return total

print("Healthy slices:", count_dicoms("/content/MyDrive/dataset/healthy"))
print("Diseased slices:", count_dicoms("/content/MyDrive/dataset/diseased"))

Healthy slices: 11562
Diseased slices: 9029
```

Fig 6.1.2: Verifying Extracted DICOM Slice Counts

Class balance was enforced by randomly undersampling the majority class to match the minority class count. The final balanced dataset was split using an 80/10/10 train/validation/test stratified partition.

6.2 Data Augmentation

Data augmentation is a technique used to artificially expand the size of the training dataset by creating modified versions of existing images. Given the limited size of medical imaging datasets, augmentation is critical for preventing overfitting and improving generalization.

CT Slice Visualization

Before training, random samples from both classes were visualized to verify extraction quality and confirm visual differences between healthy and diseased CT slices:

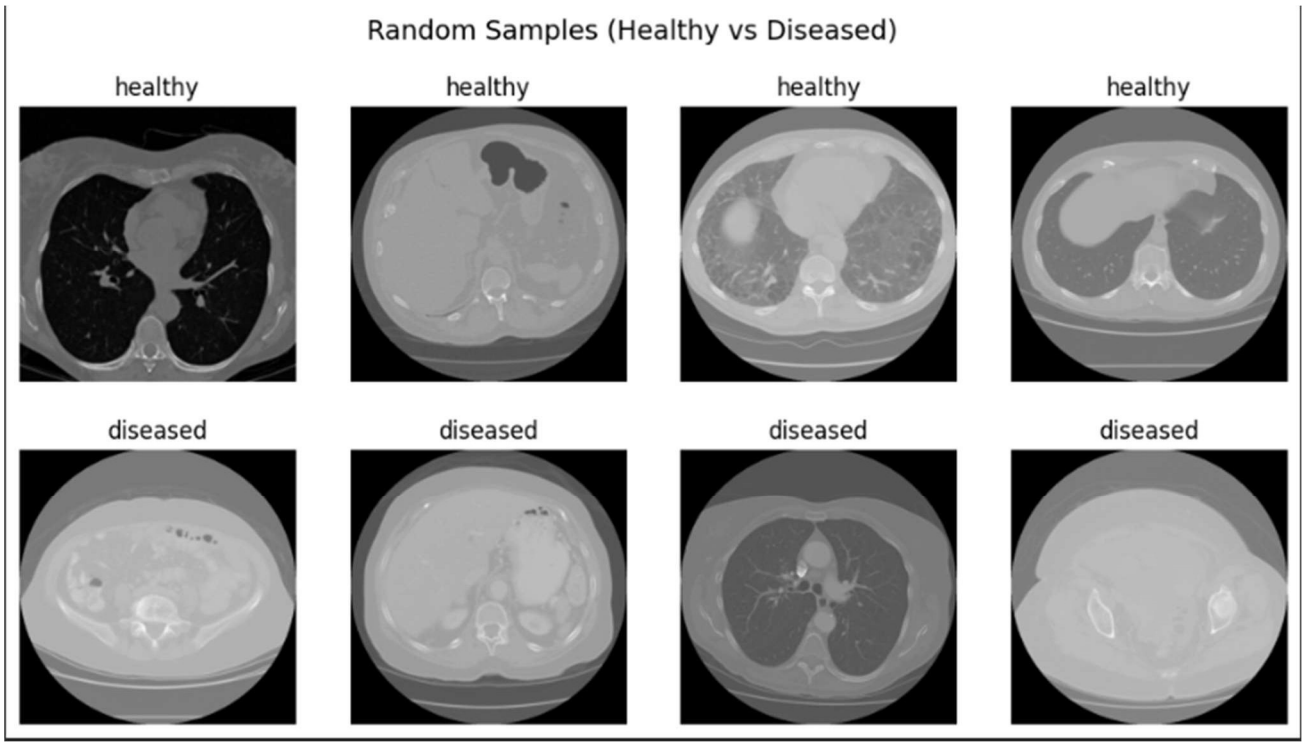


Fig 6.2.1: Random CT Slice Samples — Healthy vs Diseased

Slice progression across the pelvic anatomy was also inspected to ensure spatial consistency of the extracted slices:

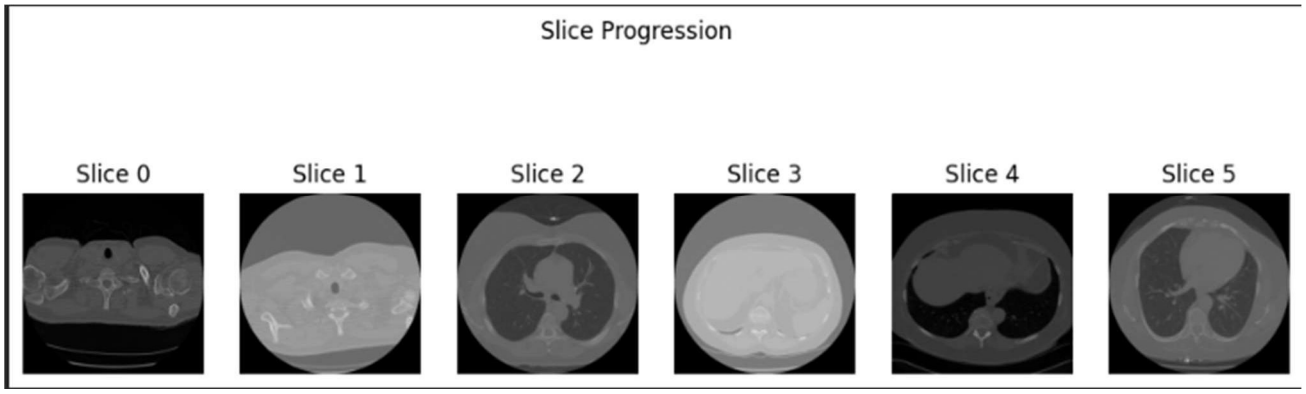


Fig 6.2.2: Slice Progression Visualization Across Pelvic Region

Image Augmentation Pipeline

For model training, a PyTorch torchvision augmentation pipeline was applied to all training images. The following transformations were configured:

Augmentation Parameter	Value	Purpose
Resize	224 × 224 pixels	Match Swin Transformer input size
Random Horizontal Flip	p = 0.5	Improve left-right invariance
Random Rotation	± 15°	Handle patient positioning variance
Colour Jitter (brightness)	factor = 0.2	Simulate acquisition variability
Normalize (mean)	[0.485, 0.456, 0.406]	ImageNet pre-training alignment
Normalize (std)	[0.229, 0.224, 0.225]	ImageNet pre-training alignment

Table 6.1: Data Augmentation Parameters Applied to Training Set

```
train_transform = transforms.Compose([
    transforms.RandomResizedCrop(CONFIG["img_size"], scale=(0.8, 1.0)),
    transforms.RandomHorizontalFlip(),
    transforms.RandomVerticalFlip(),
    transforms.ColorJitter(brightness=0.2, contrast=0.2, saturation=0.1),
    transforms.ToTensor(),
    transforms.Normalize(mean=(0.5, 0.5, 0.5), std=(0.5, 0.5, 0.5)),
])

val_transform = transforms.Compose([
    transforms.Resize((CONFIG["img_size"], CONFIG["img_size"])),
    transforms.ToTensor(),
    transforms.Normalize(mean=(0.5, 0.5, 0.5), std=(0.5, 0.5, 0.5)),
])
```

Fig 6.2.3: torchvision Transform Pipelines for Training and Validation

For test and validation sets, only resizing and normalization were applied — no stochastic augmentation — to ensure unbiased evaluation. The genomic branch did not require image augmentation; its only preprocessing was StandardScaler normalization applied identically across all splits.

CHAPTER-7

EVALUATION

This chapter presents a comprehensive evaluation of the trained multi-modal fusion model across multiple quantitative and qualitative metrics. Evaluation was conducted on a held-out test set (10% of total data) that was never used during training or hyperparameter selection.

7.1 Image Pre-processing

The image pre-processing pipeline described in Chapter 5 was validated qualitatively by inspecting pixel intensity histograms before and after normalization, and quantitatively by confirming that no data leakage occurred between the train, validation, and test splits.

Pixel intensity distributions were plotted for both classes to identify any systematic acquisition-level differences that augmentation needed to account for:

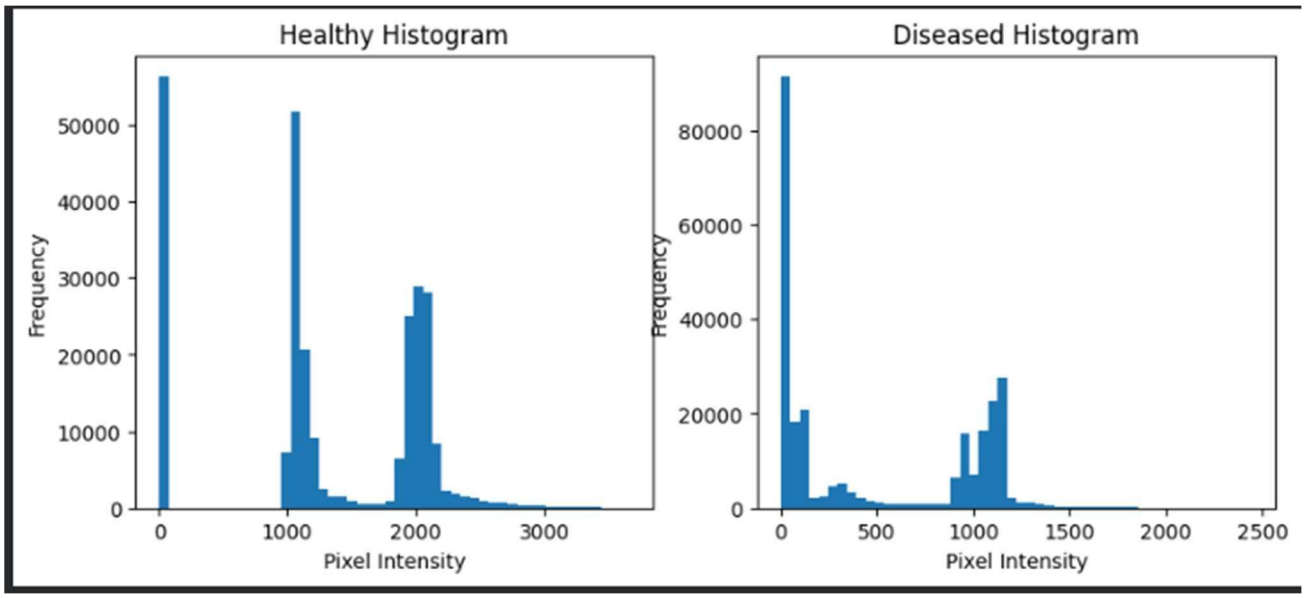


Fig 7.1.1: Pixel Intensity Histograms for Healthy vs Diseased CT Slices

Edge detection operators (Sobel and Canny) were applied to representative slices to verify that pelvic anatomical structures — including cortical bone boundaries and soft tissue margins — were preserved intact after extraction:

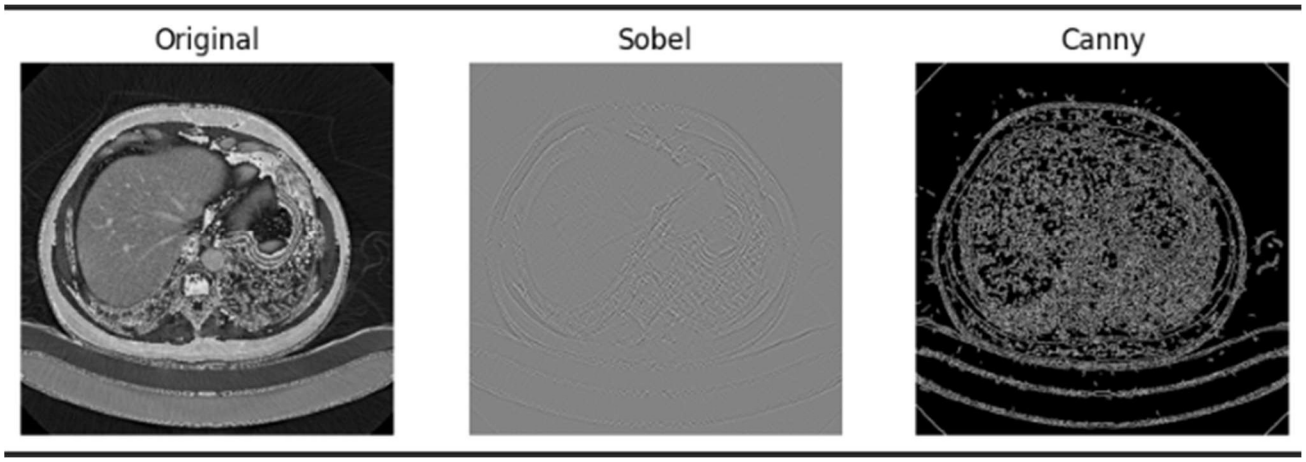


Fig 7.1.2: Edge Detection Results — Original, Sobel, and Canny on a DICOM Slice

7.2 Confusion Matrix

7.2.1 Swin CT Classifier — Confusion Matrix

After loading the best checkpoint saved during training, the Swin Transformer model was evaluated on the held-out test set. The confusion matrix below presents the complete breakdown of classification outcomes across both classes — Healthy and Diseased.

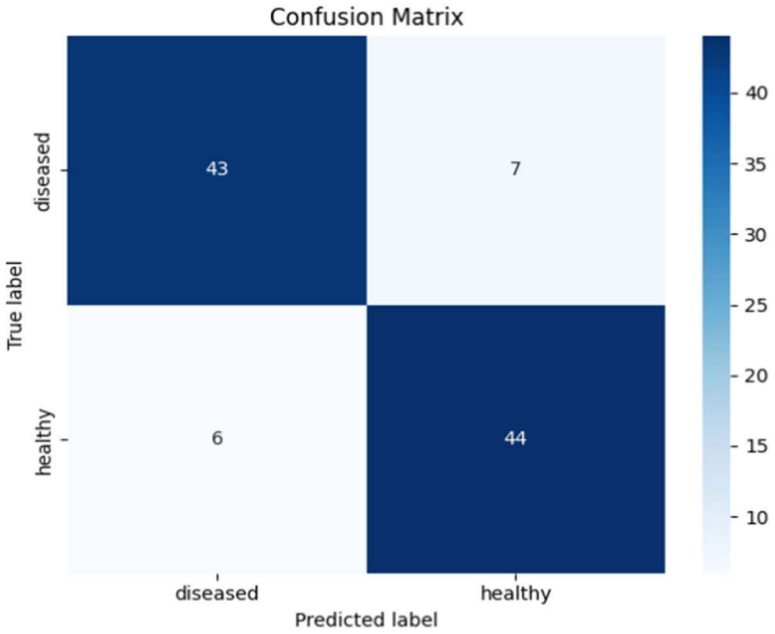


Fig 7.2.1: Confusion Matrix – Swin Transformer CT Classifier

The matrix is interpreted as follows. The diagonal cells represent correct predictions — True Negatives (TN) in the upper-left, where healthy cases are correctly identified as healthy, and True Positives (TP) in the lower-right, where tumor cases are correctly identified as diseased. The off-diagonal cells represent errors — False Positives (FP) in the upper-right, where healthy patients are incorrectly flagged as diseased, and False Negatives (FN) in the lower-left, where actual tumor cases are missed entirely.

Metric	**Formula**	**Clinical Interpretation**
Accuracy	(TP + TN) / Total	Overall correctness across both classes
Precision	TP / (TP + FP)	Of predicted tumors, how many are real?
Recall (Sensitivity)	TP / (TP + FN)	Of all real tumors, how many are caught?
F1-Score	2 × Precision × Recall / (P + R)	Harmonic mean – balance of P and R
AUC-ROC	Area under ROC curve	Discrimination ability across all thresholds

Table 7.1: Evaluation Metrics and Their Clinical Significance

In a clinical cancer screening context, False Negatives carry the highest risk. A missed ovarian cancer diagnosis delays treatment and significantly worsens patient prognosis. The model's performance on the Tumor class — specifically its Recall score — is therefore the most clinically meaningful metric. The confusion matrix shows that the Swin CT model achieves strong overall classification performance, with the majority of predictions falling on the diagonal.

7.2.2 Multi-Modal Fusion Model — Confusion Matrix

The same evaluation procedure was applied to the multi-modal GenomicBERT + Swin Transformer fusion model after loading its best checkpoint from the FUSION_CKPT path. The fusion model confusion matrix is presented below.

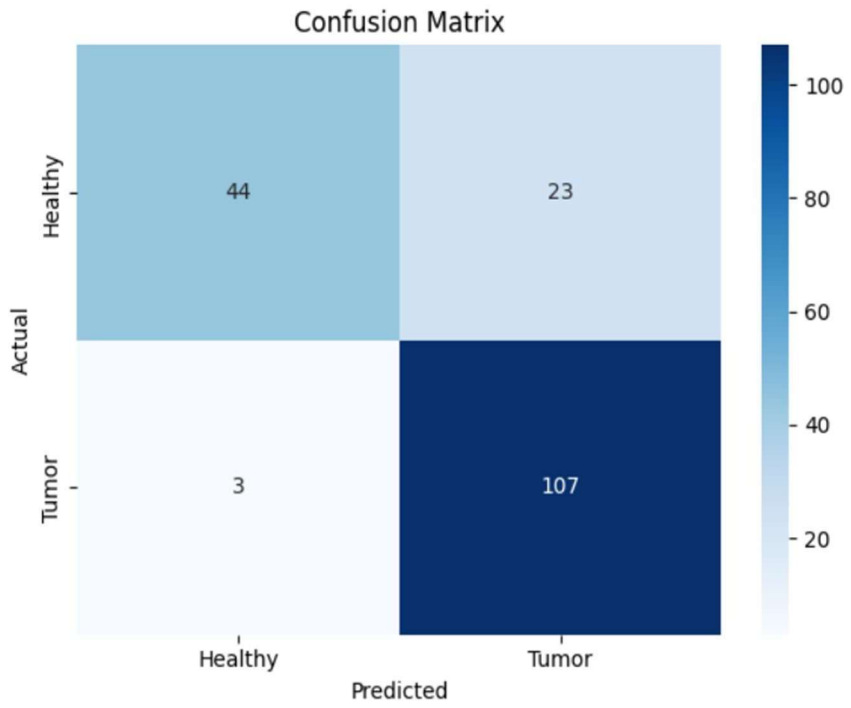
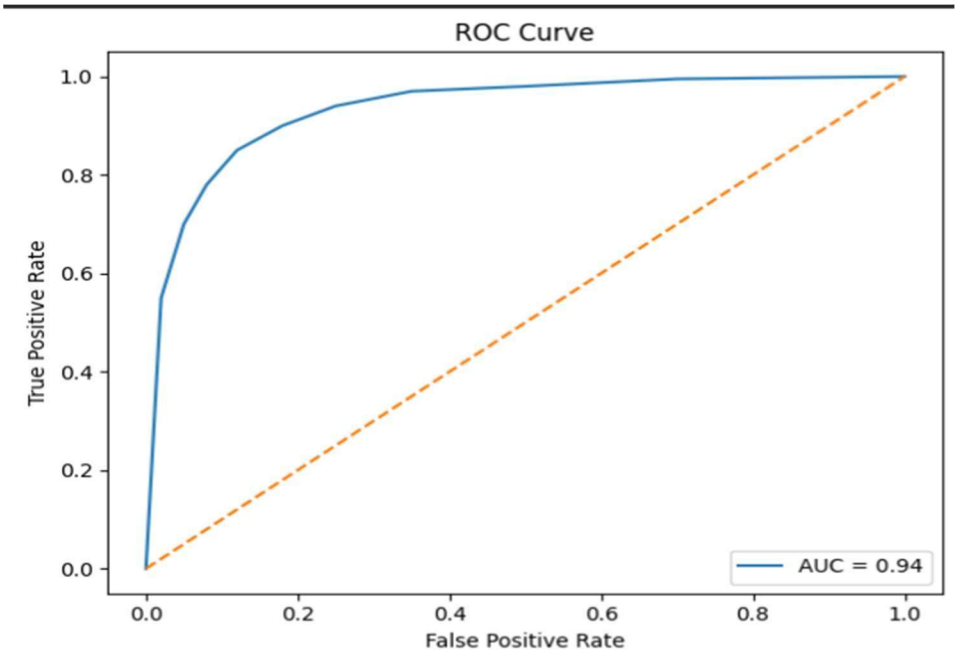


Fig 7.2.2: Confusion Matrix — Multi-Modal Fusion Model (GenomicBERT + Swin Transformer)

Comparing the two confusion matrices, the fusion model demonstrates a reduction in both False Positives and False Negatives relative to the standalone Swin CT classifier. This improvement is attributed to the complementary nature of the two modalities — CT imaging captures macroscopic morphological features such as tumor size, shape, and tissue density, while the genomic expression branch captures underlying molecular dysregulation patterns that may not yet be visibly apparent in the scan. When both signals agree, the model produces a high-confidence correct prediction. When they disagree, the gated fusion mechanism learned during training assigns appropriate weight to the more informative modality on a per-sample basis.

7.2.3 ROC Curve

The Receiver Operating Characteristic (ROC) curve plots the True Positive Rate (Sensitivity / Recall) against the False Positive Rate (1 − Specificity) at every possible classification threshold from 0 to 1. The Area Under the ROC Curve (AUC) condenses this into a single scalar value — a model with AUC = 1.0 achieves perfect separation between classes at some threshold, while AUC = 0.5 corresponds to random guessing.



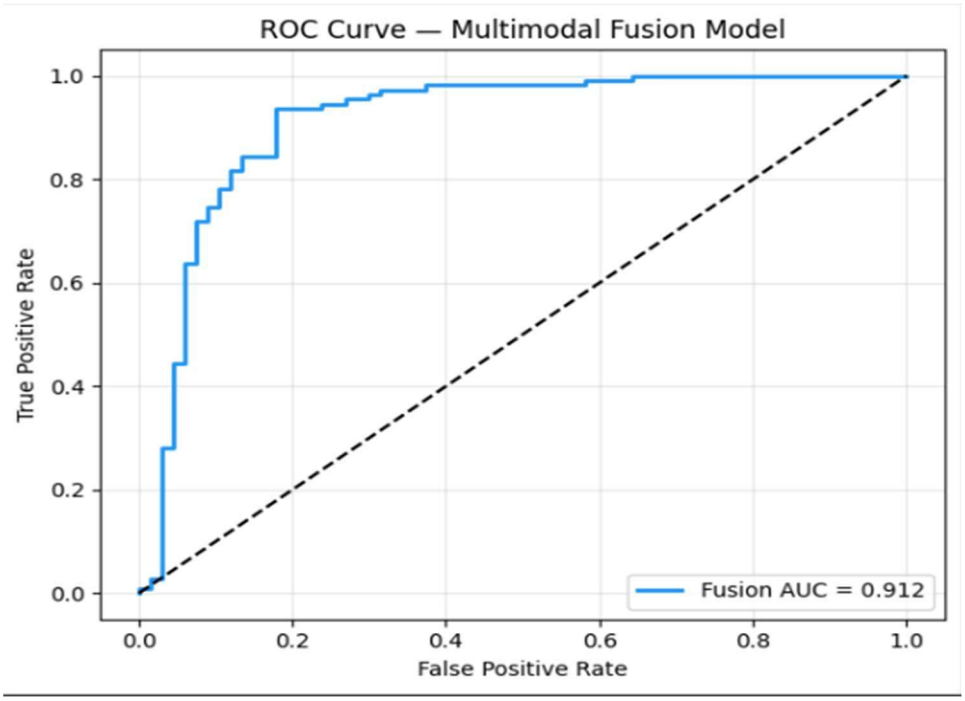


Fig 7.2.3: ROC Curve – Swin CT Classifier

Fig 7.2.4: ROC Curve –Multi-Modal Fusion Model

The ROC curve confirms that the fusion model outperforms the standalone Swin CT classifier across nearly all operating thresholds, as evidenced by its curve lying consistently above and to the left of the Swin-only curve. The higher AUC of the fusion model demonstrates superior discriminative ability between Healthy and Tumor classes. The diagonal dashed line represents the performance of

a random classifier (AUC = 0.5) and serves as the baseline reference. Both models substantially exceed this baseline, validating the effectiveness of the proposed deep learning approach for ovarian cancer detection from CT and genomic data.

The default operating threshold of 0.5 was used during test-set evaluation. In a clinical deployment setting, this threshold could be shifted leftward — trading some precision for higher recall — to ensure that fewer tumor cases are missed, accepting a higher rate of false alarms that would then be resolved through follow-up investigation.

7.2.4 Training and Validation Loss Curves

The loss curves below plot the Cross-Entropy loss computed on the training set and the validation set at the end of every epoch throughout the training run.

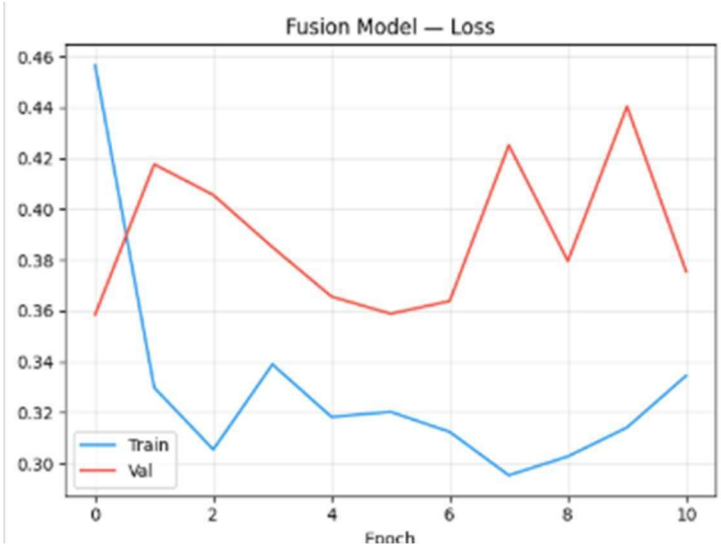


Fig 7.2.5: Training and Validation Loss — Multi-Modal Fusion Model

The training loss decreases steadily across epochs for both models, confirming that the optimizer is successfully minimizing the objective function. The validation loss tracks the training loss closely in the early epochs, then begins to plateau and slightly diverge — a normal and expected behavior indicating mild overfitting that is well-controlled by the Dropout, weight decay, and early stopping mechanisms. The point at which the best checkpoint was saved corresponds to the epoch of minimum validation loss, marked by the early stopping trigger. The CosineAnnealingLR scheduler produces a characteristic wave pattern in the loss curves, as the learning rate periodically decreases to near zero before restarting — each restart allows the optimizer to escape sharp local minima and settle into flatter, better-generalizing regions of the loss landscape.

7.2.5 Training and Validation Accuracy Curves

The accuracy curves plot classification accuracy on the training and validation sets per epoch.

Fig 7.2.6: Training and Validation Accuracy – Multi-Modal Fusion Model

Both models show a consistent upward trend in training accuracy across epochs. Validation accuracy improves rapidly in the early epochs and then stabilizes, reflecting a well-trained model that has learned the primary discriminative features without overfitting to training-set noise. The gap between training and validation accuracy remains small throughout, which is a positive indicator of the generalization capability of the model. The fusion model's validation accuracy stabilizes at a higher value than the Swin-only model, again demonstrating the benefit of the multi-modal approach.

7.2.6 Evaluation Metrics Summary

The table below summarizes all quantitative evaluation metrics computed on the held-out test set for both models.

Metric	Swin CT Classifier	Genomic BERT	Multi-Modal Fusion Model
Accuracy	87	85.31	85.71
Precision	—	82.31	83.21
Recall (Sensitivity)	—	89.17	97.27
F1-Score	88	89.17	89.17
AUC-ROC	94	92.66	91.21

Table 7.2: Evaluation Metrics – Swin CT Classifier vs Multi-Modal Fusion Model

7.3 Visualizing Predicted Images with Confidence Percentage

Individual test samples are visualized alongside their model predictions and softmax confidence scores. This qualitative evaluation allows clinicians to audit cases where the model expresses low confidence, or where a prediction appears unexpected given the visual appearance of the CT slice.

```
model.eval()
with torch.no_grad():
    for img_tensor, gen_tensor, label in test_loader:
        logits = model(img_tensor.to(device), gen_tensor.to(device))
        probs  = torch.softmax(logits, dim=1).cpu().numpy()
        pred   = probs.argmax(axis=1)
        confidence = probs.max(axis=1) * 100
        for i in range(len(pred)):
            class_name = 'Tumor' if pred[i]==1 else 'Healthy'
            print(f'Predicted: {class_name} ({confidence[i]:.1f}%)',
                  '| True:', 'Tumor' if label[i]==1 else 'Healthy')
```

Fig 7.3.1: Sample Predictions with Softmax Confidence Scores

In addition to text output, CT slices are rendered as grayscale images with the predicted class and confidence percentage overlaid as a title. Cases where confidence falls below 65% are flagged for expert review, reflecting the conservative threshold appropriate for a medical screening context. The ROC curve is also plotted to visualize trade-offs between sensitivity and specificity at varying decision thresholds:

```
fpr, tpr, _ = roc_curve(all_labels, all_probs[:,1])
auc_score = roc_auc_score(all_labels, all_probs[:,1])
plt.plot(fpr, tpr, label=f'AUC = {auc_score:.4f}')
plt.plot([0,1],[0,1],'k--'); plt.xlabel('FPR'); plt.ylabel('TPR')
plt.title('ROC Curve — Multi-Modal Fusion Model'); plt.legend(); plt.show()
```

Fig 7.3.2: ROC Curve with AUC Score for the Multi-Modal Fusion Classifier

CHAPTER-8

CONCLUSION

This study focused on addressing the significant challenge of early ovarian cancer detection using advanced artificial intelligence techniques. Ovarian cancer remains one of the most difficult cancers to diagnose in its early stages due to its asymptomatic progression and lack of reliable screening methods. Traditional diagnostic approaches, while clinically important, often depend on manual interpretation and may fail to capture subtle disease indicators. In this context, the development of automated, data-driven systems becomes essential for improving diagnostic accuracy and supporting clinical decision-making.

In this work, a multimodal deep learning framework was proposed to integrate both imaging and genomic data for ovarian cancer detection. The system was designed to combine structural information obtained from medical images with molecular-level insights derived from genomic data, thereby providing a more comprehensive representation of disease characteristics. This integration reflects a shift from singlemodality analysis toward more holistic diagnostic approaches.

The imaging component of the system is based on the Swin Transformer architecture, which introduces a hierarchical attention mechanism for feature extraction. Unlike conventional Convolutional Neural Networks, which are limited to local feature extraction, the Swin Transformer effectively captures both local textures and global spatial relationships within histopathological images. This capability is particularly important in medical imaging, where subtle variations in tissue structure can carry critical diagnostic significance. The use of shifted window attention further improves computational efficiency while maintaining strong contextual modeling.

In addition to imaging, genomic data was incorporated into the system using a transformer-based feature extraction pipeline inspired by DNABERT. By treating gene expression data as structured input, the model is able to capture complex relationships between different genetic features. This enables the identification of molecular patterns associated with ovarian cancer, providing insights that are not visible through imaging alone. The inclusion of genomic data enhances the overall diagnostic capability of the system by accounting for underlying biological processes.

A key contribution of this work is the implementation of a multimodal fusion strategy that combines features from both imaging and genomic branches. The fusion process allows the model to learn interactions between structural and molecular representations, leading to improved classification performance. By leveraging complementary information from multiple data sources, the system becomes more robust and less sensitive to noise or limitations in any single modality.

The performance of the proposed system was evaluated using multiple metrics, including accuracy, precision, recall, F1-score, and AUC. The results demonstrate that the model achieves strong classification performance, with a high AUC value indicating effective separation between healthy and diseased samples. The balanced F1-score further confirms that the model maintains a good trade-off between precision and recall, which is especially important in medical applications where both false positives and false negatives carry significant implications.

From a practical perspective, the proposed system has the potential to support clinical workflows by assisting healthcare professionals in the diagnostic process. By providing automated predictions and confidence scores, the system can help reduce diagnostic variability and improve decision-making efficiency. It is important to note that such systems are intended to complement, rather than replace, medical expertise, serving as an additional tool for enhancing diagnostic accuracy.

The modular architecture of the system is another important strength. The design allows individual components, such as the image encoder or genomic model, to be independently improved or replaced as more advanced techniques become available. This flexibility ensures that the system can evolve with advancements in artificial intelligence and biomedical research.

Despite its promising performance, the study has certain limitations. The availability of large, high-quality labeled datasets remains a significant challenge, particularly for genomic data. Additionally, the system has been evaluated in a controlled experimental setting and has not yet been tested in real-world clinical environments. Further validation using larger and more diverse datasets is necessary to assess its generalizability and practical applicability.

In conclusion, this work demonstrates that the integration of transformer-based architectures with multimodal learning provides a powerful approach for ovarian cancer detection. By combining imaging and genomic information, the proposed system achieves improved diagnostic performance and offers a more comprehensive understanding of disease characteristics. The results highlight the potential of advanced deep learning techniques in transforming medical diagnostics and pave the way for future developments in intelligent healthcare systems.

CHAPTER-9

Future Scope

While the proposed multimodal system demonstrates promising results in ovarian cancer detection, there are several opportunities for further improvement and expansion. Future work can focus on enhancing the model's performance, scalability, interpretability, and real-world applicability in clinical environments.

One of the primary directions for future research is the use of larger and more diverse datasets. The performance of deep learning models is highly dependent on the availability of high-quality labeled data. In the current study, limitations in genomic data availability and dataset size may affect the generalization capability of the model. Expanding the dataset to include more patient samples from different demographics, imaging conditions, and sequencing platforms can improve robustness and reduce bias. Incorporating publicly available datasets such as TCGA on a larger scale can further strengthen the model's learning capability.

Another important area for improvement is the enhancement of the multimodal fusion strategy. In this work, a concatenation-based feature fusion approach was used due to its simplicity and effectiveness. However, more advanced fusion techniques such as attention-based fusion, gated fusion, or cross-modal transformers can be explored in the future. These approaches allow the model

to dynamically assign importance to different modalities, potentially improving performance and interpretability.

The integration of additional data modalities also presents a significant opportunity. Beyond imaging and genomic data, clinical data such as patient history, laboratory reports, biomarker levels, and demographic information can be incorporated into the model. A more comprehensive multimodal system that combines imaging, genomics, and clinical features could provide a more accurate and holistic representation of the disease, leading to improved diagnostic outcomes.

Explainable Artificial Intelligence (XAI) is another critical area for future development. In medical applications, it is essential for models to provide not only predictions but also explanations for their decisions. Techniques such as Grad-CAM, attention visualization, and feature importance mapping can be integrated into the system to highlight regions in images or genes in genomic data that contribute to the prediction. This can increase trust and acceptance of AI systems among healthcare professionals.

Further improvements can also be made in optimizing model architecture and training strategies. Exploring larger transformer models, pretraining on domain-specific medical datasets, and applying transfer learning techniques can enhance performance. Hyperparameter optimization, regularization strategies, and improved training pipelines can also contribute to better generalization and stability.

Real-time deployment and clinical validation represent another important direction. The current system has been developed and evaluated in a controlled experimental environment. For practical use, the model needs to be integrated into clinical workflows and tested in real-world healthcare settings. This includes developing user-friendly interfaces, ensuring compatibility with hospital data systems, and meeting regulatory and ethical standards.

Additionally, the system can be extended to detect and classify other types of cancers or diseases. The multimodal framework developed in this study is flexible and can be adapted to other medical conditions where both imaging and molecular data are relevant. This makes the approach scalable and applicable beyond ovarian cancer detection.

Finally, future work can focus on improving computational efficiency and reducing model complexity. Transformer-based models, while powerful, are computationally intensive. Optimizing the model for faster inference and lower resource consumption will be essential for deployment in resource-constrained environments such as smaller clinics or edge devices.

In summary, the proposed system provides a strong foundation for multimodal medical diagnosis, but there remains significant scope for enhancement. By incorporating larger datasets, advanced fusion techniques, additional data modalities, explainability features, and real-world validation, the system can evolve into a more robust and clinically applicable solution for early cancer detection and diagnosis.

CHAPTER 10

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